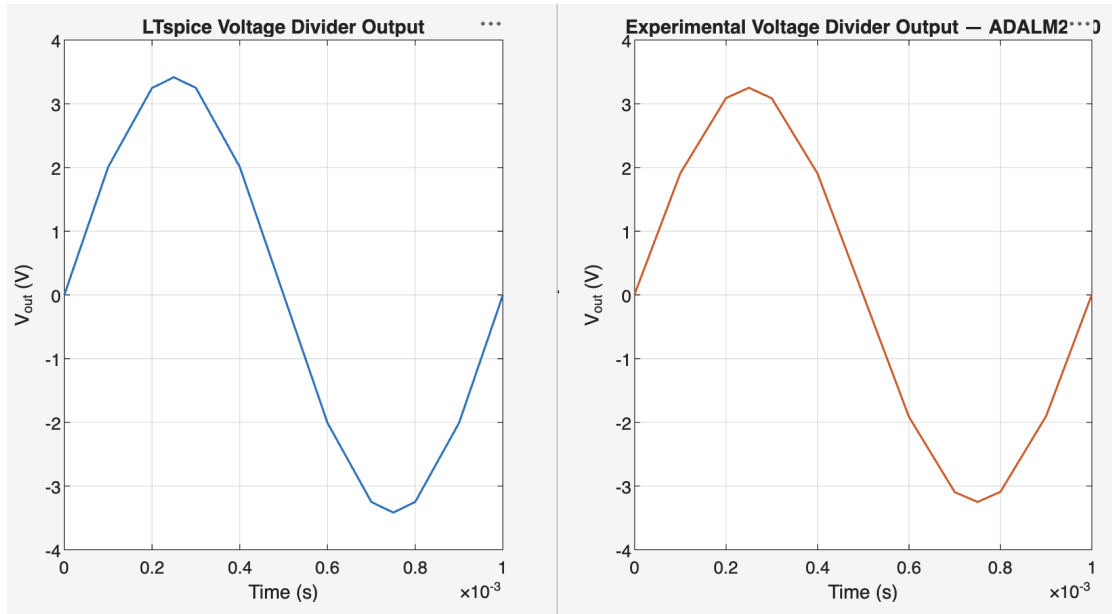


Integration Comparison

Same voltage-divider idea compared across hand calculation, LTspice/MATLAB, and the ADALM2000 measurement.

Hand calculation: $V_{out} = 5 \times [1100 / (510 + 1100)] = 3.416 \text{ V}$.



MATLAB comparison of simulated and experimental divider output.



ADALM2000 / Scopy measurement: $V_{out} = 3.251 \text{ VDC}$ for a +5 V input.

The measured value is slightly below the ideal calculation. Resistor tolerance, generator accuracy, and breadboard contacts can cause this difference.